

BRENNER, Austria

WMO: 111280

Lat: 47.0072N

Lon: 11.5108E

Elev: 1447

StdP: 85.11

Time Zone: 1.00 (EUC)

Period: 94-08

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.8	-12.1	-17.0	1.0	-11.9	-15.5	1.2	-10.4	8.6	-3.2	6.8	-3.0	1.3	90	0.623

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.7	24.1	15.1	22.4	14.4	20.7	13.6	16.0	22.2	15.2	20.5	14.4	19.1	2.6	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.9	11.9	17.1	13.2	11.3	16.4	12.5	10.8	15.8	50.2	22.3	47.6	20.7	45.3	19.4	19.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
7.1	5.2	4.3	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			-16.6	26.7	3.2	1.3	-18.9	27.7	-20.8	28.4	-22.6	29.2	-24.9	30.2
			-16.8	17.9	3.1	1.1	-19.0	18.6	-20.8	19.2	-22.5	19.8	-24.7	20.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	4.9	-4.1	-2.9	0.2	3.9	9.1	12.2	13.8	13.3	9.8	6.2	0.1	-3.0
	DBStd	7.49	3.92	4.06	4.45	4.12	3.60	4.11	3.31	3.35	3.56	3.71	4.17	3.79
	HDD10.0	2257	436	360	304	185	61	26	8	9	46	123	298	402
	HDD18.3	4894	694	593	562	432	286	186	142	159	256	374	548	660
	CDD10.0	410	0	0	0	3	33	92	126	111	40	6	0	0
	CDD18.3	6	0	0	0	0	0	1	2	2	0	0	0	0
	CDH23.3	68	0	0	0	0	1	18	27	22	1	0	0	0
	CDH26.7	1	0	0	0	0	0	0	0	1	0	0	0	0
Wind	WSAvg	1.6	1.4	1.5	1.7	1.7	1.8	1.8	1.7	1.4	1.4	1.4	1.4	1.3
Precipitation	PrecAvg	1089	52	43	57	71	107	136	151	156	102	89	77	48
	PrecMax	1366	103	116	161	135	173	194	243	245	156	176	194	104
	PrecMin	882	11	10	18	7	42	73	72	78	31	28	1	1
	PrecStd	128	24	25	31	35	37	35	41	42	33	38	47	26
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.0	7.6	12.6	17.7	22.7	25.5	26.0	26.1	22.1	16.2	10.4	5.5
		MCWB	2.3	3.0	6.0	9.9	13.6	15.8	16.2	16.3	13.9	11.0	7.8	2.7
	2%	DB	3.1	5.4	10.4	14.7	20.0	23.6	23.9	23.5	19.2	14.4	8.0	3.6
		MCWB	0.5	1.5	4.8	8.2	12.2	14.7	15.4	14.9	12.4	10.2	5.9	1.8
	5%	DB	1.8	3.5	8.2	12.3	17.7	21.5	21.9	21.5	17.2	12.6	6.4	2.5
		MCWB	0.0	0.4	3.6	6.7	10.8	13.8	14.6	13.9	11.5	9.1	5.0	0.9
	10%	DB	0.7	2.1	6.2	10.3	15.4	19.5	20.0	19.4	15.4	11.1	5.3	1.5
		MCWB	-0.6	-0.2	2.6	5.6	9.8	13.0	13.6	13.5	11.0	8.5	4.2	0.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.6	3.6	6.6	10.4	14.1	16.8	17.2	17.1	15.0	12.0	8.8	3.9
		MCDB	4.5	6.7	11.6	17.2	21.7	23.3	23.7	24.4	19.6	14.5	9.9	4.4
	2%	WB	1.3	2.1	5.2	8.6	12.6	15.5	16.0	15.8	13.6	11.0	6.4	2.3
		MCDB	2.3	4.3	9.3	13.6	18.6	21.6	22.1	21.6	17.8	13.2	7.4	3.0
	5%	WB	0.5	1.0	4.1	7.3	11.4	14.5	15.2	14.7	12.5	9.9	5.2	1.4
		MCDB	1.3	2.7	7.4	11.2	16.6	19.9	20.4	19.5	15.7	11.9	6.3	2.1
	10%	WB	-0.4	0.1	3.1	6.1	10.3	13.6	14.4	14.0	11.5	8.8	4.3	0.5
		MCDB	0.5	1.7	5.5	9.3	14.4	18.4	18.8	18.1	14.5	10.8	5.2	1.2
Mean Daily Temperature Range	5% DB	MDBR	5.0	6.4	7.8	7.9	9.0	9.4	9.7	8.8	8.5	6.8	4.7	4.5
		MCDBR	5.5	8.2	10.6	11.8	12.9	13.2	12.6	12.5	11.6	8.8	5.7	4.9
		MCWBR	4.1	5.5	5.9	5.8	5.6	5.4	5.1	5.2	5.5	5.2	3.9	3.7
	5% WB	MCDBR	4.9	7.1	9.5	10.8	11.5	12.0	11.5	11.1	10.4	7.6	4.9	4.1
		MCWBR	4.0	5.2	5.6	5.3	5.3	5.3	5.3	5.3	5.4	4.7	3.6	3.3
Clear-Sky Solar Irradiance	taub	0.240	0.264	0.303	0.346	0.360	0.368	0.365	0.359	0.329	0.305	0.273	0.241	
	taud	2.371	2.306	2.277	2.305	2.339	2.357	2.361	2.397	2.466	2.507	2.513	2.437	
	Ebn at Noon	874	914	920	908	904	895	892	884	886	859	829	834	
	Edh at Noon	69	92	111	120	121	119	117	109	92	76	61	58	
All-Sky Solar Radiation	RadAvg	1.48	2.42	3.57	4.39	4.85	5.49	5.50	4.84	3.67	2.50	1.49	1.22	
	RadStd	0.17	0.29	0.41	0.59	0.54	0.49	0.47	0.48	0.41	0.30	0.23	0.16	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (91 neighbors)	+0.37	N/A	N/A	+0.64	+0.40	+0.38	N/A	N/A	+62	+14

Nomenclature: See separate page